

Question		Marking details	Marks available				Maths	Prac
			AO1	AO2	AO3	Total		
1	(a)	$v = \frac{I}{nAe}$ correct transposition at any stage or by implication(1) $A = 12.00 \times 10^{-3} \text{ m} \times 2.00 \times 10^{-3} \text{ m} [= 2.4 \times 10^{-5} \text{ m}^2]$ (1) $v = 2.45 \times 10^{-5} [\text{m s}^{-1}]$ (1) Accept $2.5 \times 10^{-5} [\text{m s}^{-1}]$		3		3	3	
	(b)	Energy = $I^2 R t$ or equivalent used (1) 192 J [or 190 J] UNIT (1) Collisions between [free] <u>electrons and ions</u> [or atoms or lattice] (1)	1	1 1		3	2	
		Question 1 total	1	5	0	6	5	0